

Re-Os (ICP-MS) dating of the massive sulfide ores from the Jinchuan Ni-Cu-PGE deposit

YANG Gang^{1,2}, DU Andao¹, LU Jiren³, QU Wenjun¹ & CHEN Jiangfeng²

1. National Research Center of Geoanalysis, Beijing 100037, China;

2. Laboratory for Earth Material Cycling, School of Earth and Space Sciences, University of Science and Technology of China, Hefei 230026, China;

3. Institute of Mineral Resource, Chinese Academy of Geological Sciences, Beijing 100037, China

Correspondence should be addressed to Yang Gang (email: ygang@mail.ustc.edu.cn)

Received December 15, 2003; revised August 23, 2004

Abstract Massive sulfide ore from the Jinchuan Ni-Cu-PGE deposit was dated by Re-Os technique using Carius Tube digestion, Os distillation, Re extraction by acetone and ICP-MS measurement. An isochron giving an age of 833 ± 35 Ma, initial $^{187}\text{Os}/^{188}\text{Os}$ ratio of 0.279 ± 0.018 (MSWD = 1.7) and $\gamma_{\text{Os}} = 130 \pm 15$ is obtained. The Re-Os age obtained agrees, within analytical uncertainties, with previous SHRIMP zircon U-Pb age of 827 ± 8 Ma ($n=5$, MSWD = 4.3). The result suggests that the deposit was formed at Neoproterozoic and that the formation of the Jinchuan Ni-Cu-PGE sulfide deposit may be closely related to the plume activity, which were attributed to the breakup of the Rodinia supercontinent. The high γ_{Os} of the massive ores suggests that crustal material was involved in the formation of the Jinchuan deposit, however, the quantity of the added crustal material is less than 10%.

Keywords: Jinchuan Ni-Cu-PGE deposit, Re-Os dating, ICP-MS.

DOI: 10.1360/02yd0124

The Jinchuan Ni-Cu-PGE deposit is the largest nickel sulfide deposit in China and the third largest in the world^[1]. It is also the largest platinum deposit in China and contains about 60% platinum reserve of China^[2]. The Jinchuan intrusion covers only an area of 1.34 km^2 , but 47.8% of its volume is the orebody. That makes Jinchuan a typical “large ore deposit in a small intrusion”. A number of research works has been carried out towards the Jinchuan deposit. Different ore-forming models have been postulated, they include: (1) the residual root zone of a huge dyke (a differentiated high-magnesium basaltic magma) and (2) multiple stage injection with *in-situ* liquation. However, most models agree that the ore-forming material is derived from the upper mantle^[1,3–11].

Ore-forming age is crucial to testing ore genesis theories. Tang et al. (1991) reported a Sm-Nd isochron age of 1508 ± 31 Ma. The isochron was constructed by lherzolite, pyroxene-peridotite, plagioclase lherzolite and their constitute minerals^[12]. Li et al. (2004) reported a SHRIMP U-Pb zircon age of 827 ± 8 Ma for the sulfide-bearing troctolite from the Jinchuan deposit^[13]. Guiyang Institute of Geochemistry, Chinese Academy of Sciences, has reported a biotite K-Ar age of 820 Ma for the Jinchuan deposit^[14]. Recently, Zhang et al. (2004) reported a Re-Os isochron age of 1043 ± 28 Ma ($n = 7$, MSWD = 1.3). The isochron was constructed by both net-textured and massive ores from the Jinchuan deposit^[15]. Because of the disagreement of ages in literature, it is necessary to make

a critical check for the ore-forming age of the Jinchuan deposit.

The Re-Os isotopic system possesses a unique characteristic by which the system can be used to directly date metal sulfide ore and trace the source of the ore-forming material. Many metal sulfide deposits in China and worldwide have been successfully dated by using Re-Os isotopic system^[16–19]. In this paper, we report the Re-Os dating result of the massive sulfide ore from the Jinchuan deposit and provide an important constraint for the ore-forming model of the deposit.

1 Geologic setting

The Jinchuan Ni-Cu sulfide deposit is located in the Longshoushan uplifted terrane at the southwestern margin of the North China Craton. The Longshoushan uplifted terrane is bordered to the Central Asia orogenic belt in the north and the Qilian-Qinling orogenic belt in the south (Fig. 1). The basement of the Longshoushan region consists of Archean metamorphic rock. In the uplifted terrane, the structural lineation strikes NW-NNW, which is controlled by two deep faults on the north and south sides of the region. A typical feature of the uplifted terrane is the development of different types of faults. A series of secondary faults parallels the main faults on the north and south margins of the uplifted terrane and these secondary faults increase in number toward the main faults, forming two parallel zones. The ore-bearing intrusion is controlled by secondary northwest and northeast striking faults. The Prechangchengian Longshoushan group occurs in

the Longshoushan uplifted terrane and comprises the Baijiazui and Tamazigou formations from bottom to top. The Baijiazui formation consists mainly of homogenic migmatites, biotite-plagioclase-gneisses, chlorite-quartz-schists and serpentine marbles and the Tamazigou formation schists, gneisses and marbles. The Dunzigou formation of the Changcheng-Jixian systems, overlying on the Longshoushan group, consists of conglomerates, sandstones and crystalline limestones, representing a typical continental depositional environment^[20].

The magmatic activity in the Longshoushan uplifted terrane is abundant and various intrusions of ultrabasic to acid in composition occur in the region. The discontinuously distributed ultrabasic bodies in the region form an important mafic-ultramafic rock belt. The mafic-ultramafic rocks cropping out in the western part of the region are represented by the Zangbutai intrusion, but those in the central part by the Jinchuan intrusion and those in the eastern part by the Xiaokouzi intrusion. As an irregular dyke, the Jinchuan Ni-Cu-bearing ultramafic intrusion intruded into metamorphic rocks of the Prechangchengian Baijiazui formation. The direct wall rocks are homogenic migmatites, striped migmatites, biotite-plagioclase-gneisses, garnet two mica gneisses, chlorite-quartz-schists and serpentine marbles. The Jinchuan intrusion is about 6500 m long, 300 m wide and more than 1000 m deep in the central part. The rock body strikes NW-SE and dips to the SW. The Jinchuan intrusion and its ores have been divided into four mining areas, known formally, from west to east, as III, I, II and IV mining areas by the NEE striking F8, F16-1 and F-23 faults (Fig. 2). The orebody of the Jinchuan Ni-Cu sulfide deposit is divided into five different genetic types: (1) *in-situ* magmatic liquation orebody; (2) deep-seated magmatic liquation-injection orebody; (3) late-stage injection orebody; (4) contact metasomatic orebody; and (5) superimposed hydrothermal orebody^[14,20].

The Jinchuan intrusion contains four rock types. They are dunite, lherzolite, plagioclase lherzolite and pyroxenite. The Jinchuan Ni-Cu sulfide-bearing body is a composite intrusion resulting from multiple injec-

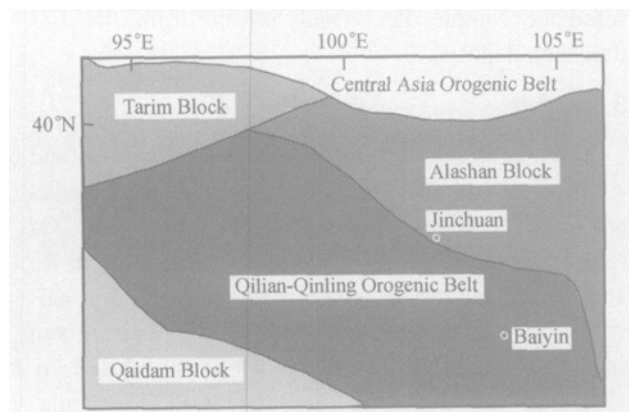


Fig. 1. Simplified regional geologic map of the Jinchuan area^[14].

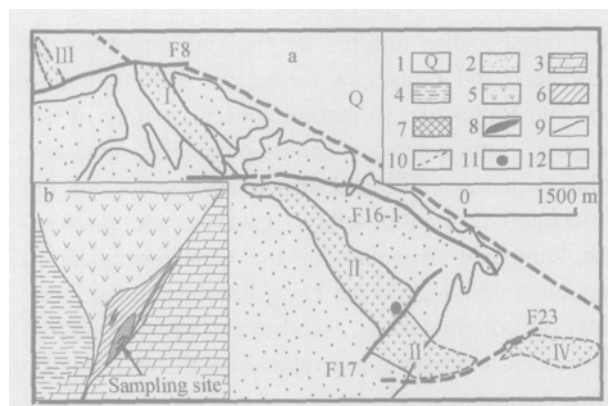


Fig. 2. A sketch of the Jinchuan Ni-Cu sulfide deposit^[20]. 1, Quaternary sediment; 2, Prechangchengian Baijiazui formation; 3, marble; 4, migmatite; 5, ultramafic rock; 6, disseminated ore; 7, net-textured ore; 8, massive sulfide ore; 9, fault; 10, deep fault; 11, sampling site; 12, mining area. a, The sketch of the Jinchuan deposit; b, a cross section of the sampling site.

tions of magma with four stages. The stage 1 intrusion includes fine- to medium-grained lherzolite, olivine pyroxenite and pyroxene-peridotite. The corresponding ore is sparsely disseminated and disseminated ore, occurring as small suspended lenses. The stage 2 intrusions comprise medium- to coarse-grained pyroxene-peridotite, lherzolite, olivine websterite, plagioclase lherzolite and websterite. The corresponding ore is mainly disseminated and semi-net-textured ore, forming thick layered and lens orebodies in the middle and lower parts of the intrusion. The stage 3 intrusions is medium-grained pyroxene-peridotite and dunite and the ore type is net-textured ore occurring as huge lenses in the bottom part of the intrusion. The stage 3 is the main mineralization period and the rocks of this type contain 85% Ni reserve of the deposit. The stage 4 massive sulfides with little silicate mineral formed veins, lenticular and chambered orebodies^[11,14,21].

Representative massive Ni-Cu sulfide ores occurring near the contact zone of the bottom part of the mining area II are chosen for Re-Os dating. The samples were formed at stage 4. Most of them are massive ores, and a few occur at the edges of the Ni-Cu sulfide orebodies as semi-massive and brecciated ores. Sulfide minerals include pyrrhotite, pentlandite, chalcopyrite, pyrite and violarite, with minor magnetite and hematite. No late alteration has been found under a micro-

scope^[14].

2 Analytical method

All the chemistry and ICP-MS measurement were performed in the National Research Center of Geoanalysis (NRCG), Chinese Academy of Geological Sciences (CAGS). Re and Os concentrations were determined by isotope dilution. The ¹⁸⁵Re and ¹⁹⁰Os spikes were originally purchased from Oak Ridge National Laboratory, USA, and calibrated at NRCG afterwards. The samples were crushed and ground to less than 200 mesh using contamination-free device. About 0.2 g sample was precisely weighed and loaded into the Carius Tube, together with ¹⁸⁵Re and ¹⁹⁰Os mixed spikes and digested by the reverse aqua regia. Os was separated as OsO₄ by distillation at 105–110°C and was trapped by Milli-Q water. Re was extracted from the residue by acetone in 5 M NaOH solution^[22].

Re and Os isotopic compositions were measured by using an Inductively Coupled Plasma Mass-spectrometer (ICP-MS, PQ Excell). The intensity of ¹⁹⁰Os signal was monitored to correct for the trace Os in Re and ¹⁸⁵Re intensity was used to monitor the trace Re in Os. 5% ammonia and H₂O₂ were repeatedly used to wash the Teflon injection tube in order to avoid the cross-contamination between different samples. The total procedural blanks are about 20 to 30 pg for Re and 1 pg for Os based on blank runs analyzed together with the samples. The ¹⁸⁷Os/¹⁸⁸Os of the blank is 0.25 based on routine measurements of blanks. The accuracy of the analysis was controlled by the laboratory reference sample JDC (molybdenite from the Jinduicheng deposit).

3 Result

Concentrations of Re and Os and ¹⁸⁷Re/¹⁸⁸Os and ¹⁸⁷Os/¹⁸⁸Os isotopic ratios corrected for total blanks are listed in Table 1. In-run precisions of ¹⁸⁷Os/¹⁸⁸Os ratios determined by ICP-MS and uncertainties in Re and Os concentrations given by isotope dilution calculation based on the error propagation are also given in Table 1. The Re-Os isochron calculated by ISOPLOT^[23] is shown in Fig. 3. In ISOPLOT calculation, the input errors in ¹⁸⁷Re/¹⁸⁸Os and ¹⁸⁷Os/¹⁸⁸Os

Table 1 Concentrations of Re and Os and $^{187}\text{Re}/^{188}\text{Os}$ and $^{187}\text{Os}/^{188}\text{Os}$ isotopic ratios of the sulfide ores from the Jinchuan Ni-Cu sulfide deposit

Sample	Re/ng · g ⁻¹	2σ	Os/ng · g ⁻¹	2σ	$^{187}\text{Re}/^{188}\text{Os}$	$^{187}\text{Os}/^{188}\text{Os}$	2σ
JII-2	44.19	0.2	3.429	0.001	62.59	1.156	0.012
JII-8	148.0	1.5	8.976	0.002	79.28	1.372	0.004
JII-10	159.0	1.6	26.93	0.02	27.73	0.6833	0.0020
JII-13	264.6	3.3	54.03	0.08	23.54	0.6002	0.0020
JII-14	138.2	1.4	32.86	0.07	20.22	0.5580	0.0050

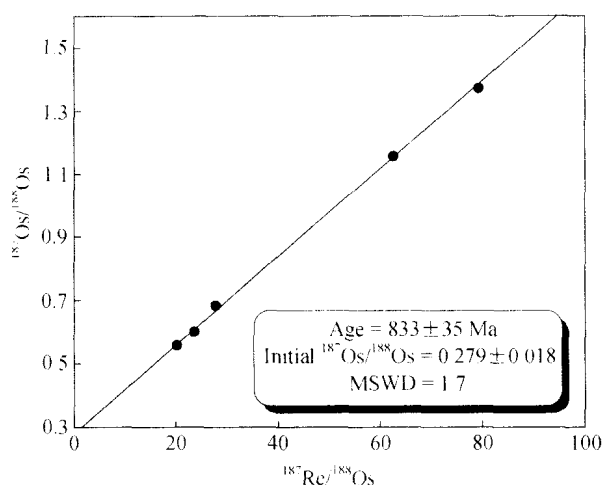


Fig. 3. Re-Os isochron for sulfide ores from the Jinchuan Ni-Cu sulfide deposit.

ratios include the uncertainties in weighing both the samples and mixed spikes, the spike calibration, the ICP-MS measurement and so on, and are represented by uncertainty of replicate analyses of the laboratory reference sample and are 1.5% for $^{187}\text{Re}/^{188}\text{Os}$ and 1% for $^{187}\text{Os}/^{188}\text{Os}$. The contribution of the uncertainty in ICP-MS measurement to the total uncertainty is relatively small. The decay constant of ^{187}Re is $1.666 \times 10^{-11} \text{a}^{-1}$ [24]. The massive sulfide ores from the Jinchuan sulfide deposit give an isochron age of $833 \pm 35 \text{ Ma}$ with an initial $^{187}\text{Os}/^{188}\text{Os}$ ratio of 0.279 ± 0.018 ($n = 5$, $\text{MSWD} = 1.7$), corresponding to $\gamma_{\text{Os}} = +130 \pm 15$.

4 Discussion

The origin of the Jinchuan Ni-Cu-PGE deposit has long been debated. Only a few geochronological works have been carried out for the deposit. The early research carried out by Tang et al. (1991) suggested that the ore deposit was formed at $1508 \pm 31 \text{ Ma}$ [12] based on a Sm-Nd isochron. The isochron was con-

structed by lherzolite, pyroxene-peridotite, plagioclase lherzolite and their constitute minerals[12]. However, the age calculation model used by Tang et al. (1991) underestimated the uncertainty of the age[25,26]. By recalculation using ISOLPOT[23], an age of $1510 \pm 140 \text{ Ma}$ is obtained. Because the range of $^{147}\text{Sm}/^{144}\text{Nd}$ ratios is small and the age uncertainty is large, the significance of the age is obscure. Guiyang Institute of Geochemistry, Chinese Academy of Sciences, has reported a biotite K-Ar age of 820 Ma for the Jinchuan deposit[14]. Li et al. (2004) reported a zircon SHRIMP U-Pb age of $827 \pm 8 \text{ Ma}$ ($n = 5$, $\text{MSWD} = 4.3$) for zircon from the sulfide-bearing troctolite of the Jinchuan deposit[13]. Zhang et al. (2004) reported a Re-Os isochron age of $1043 \pm 28 \text{ Ma}$ ($n = 7$, $\text{MSWD} = 1.3$)[15]. The isochron was constructed by both net-textured and massive ores from the Jinchuan deposit. However, the initial $^{187}\text{Os}/^{188}\text{Os}$ ratios of the six samples out of the total seven are positively correlated with reciprocals of Os concentrations that suggested heterogeneity in initial isotopic composition or a mixing relationship. Therefore, the age reported by Zhang et al. (2004) merits further scrutiny and age of the Jinchuan deposit may probably be younger than that given by Zhang et al. (2004). We obtained a new Re-Os isochron age of $833 \pm 35 \text{ Ma}$ for the massive sulfide ores. This age agrees with the zircon SHRIMP U-Pb age within analytical uncertainties. Those data suggest that the Jinchuan Ni-Cu sulfide deposit was formed at Neoproterozoic, but not Midproterozoic.

The good lineation of the Re-Os isochron given by the massive ores from Jinchuan suggests an equilibrium state of Re and Os isotopes at the beginning of crystallization and a closed system behavior during crystallization. The large range of Re/Os ratios from 4.20 to 16.49 may be due to different partition coeffi-

cients of Re and Os between monosulfide solid solution (Mss) and sulfide liquid. Experiments show that the Fe_{1-x}S Mss first crystallized from the Cu-Ni-Fe-S system at 1191°C , followed by Ni-rich and Cu-rich Mss^[27]. The most abundant Fe-S Mss in the Jinchuan deposit is pyrrhotite, the dominant Fe-Ni-S Mss is pentlandite and Fe-Cu-S Mss chalcopyrite and cubanite^[14]. Partition coefficients of Re and Os between Mss and sulfide liquid range from 2.5 to 3.8^[27]. The Re/Os ratios of residual sulfide liquid increase along with the crystallization of Mss in a closed system, that makes a large range of Re/Os ratios for the massive sulfide ores from the Jinchuan deposit and facilitates Re-Os dating.

Formation of the Jinchuan Ni-Cu sulfide deposit is related to the uplift of the crust and the formation time of the deposit is close to the time of the unconformity on the top of the Qingbaikou system, corresponding to the Jingning movement (800 Ma). Missing of the Qingbaikou system in the Longshoushan region suggests that the region had been uplifted before 800 Ma^[5]. The ore-bearing intrusion and the massive ore were injected at a tension structural environment, similar to that of a plume tectonism. The parental magma of the Jinchuan intrusion is high-Mg picrite-tholeiite magma resulting from high degree partial melting of the mantle at a high temperature ($1400\text{--}1700^\circ\text{C}$). The origin of this type of magma is often associated with the plume activity. The Re-Os isochron age of 833 ± 35 Ma for the massive sulfide ores from the Jinchuan deposit is contemporaneous with the time when the Rodinia supercontinent broke up. Li X. H. et al. (2003) and Li Z. X. et al. (2003) suggested that South China was situated in the center of the Rodinia supercontinent and upwelling of a plume beneath the South China craton at 825 Ma initiated the breakup of the supercontinent^[28,29]. Meanwhile, the North Qilian region was at a rift stage. When and how the North China craton was separated from the Rodinia supercontinent is unclear so far. It is our result that mirrors the plume activity, which resulted in breakup of the Rodinia supercontinent, in the North China craton. The plume upwelling event resulted in not only the separation of the North China craton from the Rodinia supercontinent, but also the

formation of the Jinchuan deposit.

The massive sulfide ores of the Jinchuan deposit give an initial $^{187}\text{Os}/^{188}\text{Os}$ ratio of 0.279 ± 0.018 , corresponding to a γ_{Os} of $+130 \pm 15$. The value is much higher than that of the mantle at 833 Ma ($^{187}\text{Os}/^{188}\text{Os} = 0.121$)^[30]. The high γ_{Os} suggests an addition of the crustal material during the formation of the Jinchuan deposit. This is supported by the fact that the residual wall rock is often observed in the orebody. The γ_{Os} of massive sulfide ores of Jinchuan is in agreement with that of the Pechenga deposit, Russia ($+46$ to $+251$)^[31,32]. Walker et al. (1991) estimated that the crust material added to the Pechenga deposit was less than 10%^[31]. Similar to the Pechenga deposit, the crustal material involved in the formation of the Jinchuan deposit should be in a similar small proportion. Therefore, the ore-forming material of the Jinchuan deposit is dominantly derived from the mantle, but a small amount of crustal material is involved. Lambert et al. (2000) suggested that the mafic lower crust was an important reservoir of S and metals^[33]. A small amount of material from the lower crust was added to the Jinchuan parental magma formed by high degree partial melting of the mantle or lithospheric mantle. The addition of the lower crust material caused the saturation or over-saturation of sulfides in the parental magma, and increased concentrations of metals. The sulfide liquid, separated from the parental magma, formed the sulfide ores of the Jinchuan deposit. In this regard, the addition of the lower crust material could be a key factor of the genesis of the magmatic Ni-Cu sulfide deposit.

Acknowledgements We thank Peng Zicheng and He Hongliao for assisting with mass spectrometry. We also thank the two anonymous referees for constructive advice. This work was supported by the National Natural Science Foundation of China (Grant No. 49973017).

References

1. Naldrett, A. J., Key factors in the genesis of Noril'sk, Sudbury, Jinchuan, Voisey's Bay and other world-class Ni-Cu-PGE deposits: implications for exploration, *Austral. J. Earth Sci.*, 1997, 44: 283–15.
2. Yang, X., Li, X., Yang, Z. T. et al., Pt-bearing basic and ultrabasic bodies and Pt (group) deposits in China (in Chinese), Xi'an: Xi'an Jiaotong University Press, 1993, 1–184.
3. Chai, G., Naldrett, A. J., Characteristics of Ni-Cu-PGE mineralization and genesis of the Jinchuan deposit, Northwest China,

- Econ. Geol., 1992, 87: 1475–1495.
4. Lu, J. R., Genetic model of main magmatic Cu-Ni deposits In China, *Dixueyanjiu* (in Chinese), 1993, 27: 78–83.
 5. Tang, Z. L., The main mineralization mechanism of magma sulfide deposits in China, *Acta Geological Sinica* (in Chinese), 1996, 70(3): 237–243.
 6. Liang, Y. B., Zhu, W. F., Song, G. R., Geological and geochemical characteristics of the Jinchuan platinum group element deposit of the copper-nickel type, Gansu, *Mineral Resources and Geology* (in Chinese), 1997, 11(1): 1–13.
 7. Naldrett, A. J., World-class Ni-Cu-PGE deposits: Key factors in their genesis, *Mineral. Deposita*, 1999, 34: 227–240.
 8. Xie, G. H., Wang, Y. L., Fan, C. Y. et al., Genetic mechanisms of the Jinchuan ultramafic intrusion and associate superlarge sulfide deposit, Northwest China, *Sci. China, Ser. D*, 1998, 41(Supp.): 65–73.
 9. Barnes, S. J., Tang, Z. L., Chrome spinels from the Jinchuan Ni-Cu sulfide deposit, Gansu Province, People's Republic of China, *Econ. Geol.*, 1999, 94: 343–356.
 10. Luo, Z. H., Marakushev, A. A., Paniakh, H. A. et al., The origin of copper-nickel sulfide deposits—Exemplified by Norlsk (Russia) and Jinchuan (China), *Mineral Deposits* (in Chinese), 2000, 19(4): 330–339.
 11. Yang, H. Q., On the genesis of Jinchuan copper-nickel sulfide deposit, *Bulletin of the Chinese Academy of Geological Sciences* (in Chinese), 1991, 22: 117–135.
 12. Tang, Z. L., Yang, J. D., Xu, S. J. et al., The Sm-Nd dating of Jinchuan ore-bearing ultramafic rock, *Chinese Science Bulletin* (in Chinese), 1991, 37(10): 918–920.
 13. Li, X. H., Su, L., Song, B. et al., SHRIMP U-Pb zircon age of the Jinchuan ultramafic intrusion and its geological significance, *Chinese Science Bulletin*, 2004, 49(4): 420–422.
 14. Tang, Z. L., *Ore Genesis and Geology of the Jinchuan Ni-Cu-PGE Deposit* (in Chinese), Beijing: Geological Publishing House, 1995, 157–160.
 15. Zhang, Z. Q., Du, A. D., Tang, S. H. et al., Age of the Jinchuan copper-nickel deposit and isotopic geochemical feature of its source, *Acta Geological Sinica* (in Chinese), 2004, 78(3): 359–365.
 16. Mao, J. W., Du, A. D., 982 Ma Re-Os age of copper-nickel sulfide ores in the Baotan area, Guangxi, and its geological significance, *Science in China, Series D* (in Chinese), 2001, 31: 992–998.
 17. Li, S. R., Xiao, Q. Y., Shen, J. F. et al., Rhenium-osmium isotope constraints on the age and source of the platinum mineralization in the Lower Cambrian black rock series of Hunan-Guizhou province, China, *Science in China, Series D*, 2003, 46(9): 928–940.
 18. Stein, H. J., Morgan, J. W., Schersten, A., Re-Os dating of low-level highly radiogenic (LLHR) sulfides: the Harnas gold deposit, southwest Sweden, records continental-scale tectonic events, *Econ. Geol.*, 2000, 95: 1657–1671.
 19. Arne, D. C., Bierlein, F. P., Morgan, J. W. et al., Re-Os dating of sulfides associated with gold mineralization in central Victoria, Australia, *Econ. Geol.*, 2001, 96: 1455–1459.
 20. 6th Geological Survey Party, Bureau of Geology and Mineral Resources of Gansu Province, *Geological Characteristics of the Baijiazui Cu-Ni Sulfide Deposit* (in Chinese), Beijing: Geological Publishing House, 1984.
 21. Yang, X. Z., Peng, L. G., Dong, X. Y. et al., The characteristics and geological significance magmatic intrusion in the Jinchuan copper-nickel bearing ultrabasic intrusion, Gansu Province, *Geol. Rev.* (in Chinese with English abstract), 1991, 37(1): 70–79.
 22. Du, A. D., Zhao, D. M., Wang, S. X. et al., Precise Re-Os Dating for molybdenite by ID-NTIMS with Carius Tube sample preparation, *Rock and Mineral Analysis*, 2001, 20(4): 247–252.
 23. Ludwig, K. R., *Isoplot-A Plotting And Regression Program for Radiogenic-isoplot Date*, Version 2.92, 1994.
 24. Smoliar, M. I., Walker, R. J., Morgan, J. W., Re-Os ages of group IIA, IIIA, IVA, and IVB iron meteorites, *Science*, 1996, 271: 1099–1102.
 25. Jahn, B. M., Discussion on Sm-Nd isotopic age of Precambrian-Cambrian boundary in China, *Geol. Mag.*, 1997, 134(4): 571.
 26. Turnbull, J. M., Moorbath, S., Discussion on Sm-Nd isotopic age of Precambrian-Cambrian boundary in China, *Geol. Mag.*, 1997, 134(4): 571–574.
 27. Brennan, J. M., Re-Os fractionation in magmatic sulfide melt by monosulfide solid solution, *Earth Planet. Sci. Lett.*, 2002, 199: 257–268.
 28. Li, Z. X., Li, X. H., Kinny, P. D. et al., Geochronology of Neoproterozoic syn-rift magmatism in the Yangtze Craton, South China and correlations with other continents: evidence for a mantle superplume that broke up Rodinia, *Precambrian Research*, 2003, 122: 85–109.
 29. Li, X. H., Li, Z. X., Ge, W. C. et al., Neoproterozoic granitoids in South China: crustal melting above a mantle plume at ca. 825 Ma?, *Precambrian Research*, 2003, 122: 45–83.
 30. Meisel, T., Walker, R. J., Irving, A. J. et al., Osmium isotopic compositions of mantle xenoliths: A global perspective, *Geochim. Cosmochim. Acta*, 2001, 65: 1311–1323.
 31. Walker, R. J., Morgan, J. W., Hanski, E. et al., Re-Os systematics of early Proterozoic ferropicrites, Pechenga Complex, NW Russia: evidence for ancient ¹⁸⁷Os-enriched plumes, *Geochim. Cosmochim. Acta*, 1991, 61: 3145–3160.
 32. Shirey, S. B., Walker, R. J., The Re-Os isotope system in cosmochemistry and high-temperature geochemistry, *Ann. Rev. Earth Planet. Sci.*, 1998, 26: 423–500.
 33. Lambert, D. D., Frick, L. R., Foster, J. G. et al., Re-Os isotope systematic of the Voisey's Bay Ni-Cu-Co magmatic sulfide system, Labrador, Canada: II. Implications for parental magma chemistry, ore genesis, and metal redistribution, *Econ. Geol.*, 2000, 95(4): 867–888.